

UAV Propeller Impact Testing for Injury Assessment: A Low-Speed Collision Approach

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Abstract— Propeller-induced injuries from drone collisions pose a significant concern in the field of unmanned aerial vehicle (UAV) safety. To address this issue, the development of an effective testing method for evaluating the potential harm caused by propellers upon contact with a human body is essential. In this study, we present a novel test method designed to replicate scenarios where a propeller impacts a target at a low speed, approximately 30 inches away, and at velocities of up to 20 km/h. A comprehensive crash test protocol was devised and executed, yielding a diverse set of valuable data. This research aims to contribute to the advancement of safety measures in UAV operations by providing insights into the potential risks associated with propeller impacts and supporting the development of protective measures and safety standards.

Keywords— UAVs propeller testing, safety consideration, testing protocol, safety measures.

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I. INTRODUCTION

A. Background and motivation

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have revolutionized various industries, from aerial photography and surveillance to delivery services and recreational flying. While the proliferation of drones has brought numerous benefits, it has also raised concerns about safety, particularly in cases where propellers come into direct contact with people. Such incidents can result in severe injuries, emphasizing the need for robust testing methods to assess the potential harm caused by propellers during collisions with the human body.

This study focuses on addressing the critical issue of propeller-induced injuries in UAV operations. We recognize that the safe integration of drones into our daily lives hinges on our ability to understand and mitigate the risks associated with these aerial devices. To this end, we introduce a novel testing methodology designed to evaluate the extent of injury caused by propellers in collision scenarios, specifically when a drone collides with a person's head or other body parts.

The proposed testing approach seeks to replicate real-world situations where a UAV's propeller makes contact with a human body. In these scenarios, the distance between the propeller and the target is approximately 30 inches, and the collision occurs at low speeds of up to 20 km/h. By conducting controlled crash tests under these conditions, we aim to collect a comprehensive dataset that sheds light on the potential dangers of propeller impacts, enabling us to develop informed safety measures, protective equipment, and industry standards for UAV operations.

This research endeavors to enhance our understanding of the risks associated with UAV propeller collisions and contribute to the ongoing efforts to improve safety in the rapidly expanding field of unmanned aerial vehicles.

B. Problem statement

The issue of propeller-induced injuries due to UAV-human collisions has gained prominence, and it necessitates a robust testing methodology to assess the extent of harm caused by propellers during such encounters. Current research in UAV safety predominantly focuses on accident prevention and flight control, but the evaluation of propeller-related injuries remains relatively underexplored. Consequently, a comprehensive understanding of propeller impacts on human surrogates and the effectiveness of protective equipment is lacking.

C. Research objectives

The primary objective of this study is to develop a systematic and replicable testing method that simulates real-world scenarios where UAV propellers come into contact with the human body. We aim to assess the potential harm caused by propeller impacts and gather empirical data to inform safety measures, protective gear development, and industry standards. Specifically, our research objectives include:

1. Developing a controlled testing methodology for evaluating propeller-induced injuries.
2. Conducting experiments to replicate propeller impacts at varying speeds and distances.
3. Assessing the severity of injuries on human surrogate models (head mannequins and forearm dummies).
4. Evaluating the effectiveness of protective gear in mitigating propeller-related injuries.
5. Contributing to the enhancement of safety guidelines and recommendations for UAV operators and manufacturers.

D. Significance of the study

This research holds paramount significance in the ever-expanding realm of UAV technology and its interaction with human environments. By addressing the critical issue of propeller-induced injuries, our study aims to:

1. Enhance safety protocols and industry standards for UAV operations.
2. Provide empirical data to inform the development of protective equipment.
3. Advance the understanding of potential risks associated with UAVs, particularly in scenarios involving propeller-human interactions.
4. Foster the responsible and secure integration of UAVs into various sectors, including delivery services, infrastructure inspection, and emergency response.

II. LITERATURE REVIEW

The research work related to UAVs' propeller potential harm to human and the corresponding protective measures was reported in References [1], [15], [19], [24], [25], and [29]. The authors reported their work on the safety aspect of the testing of the rotary-wing UAVs in [8]. To better understand the background of contribution, we group the references into themes to provide a comprehensive overview of the research landscape related to UAV propeller safety and testing.

A. UAV Safety and Risk Assessment

The importance of assessing safety risks and establishing safety protocols for UAVs has been a recurring theme in the literature. Researchers have explored various aspects of safety, from propeller-induced injuries to ground impact hazards and security considerations.

Afman et al. (2018) propose a new paradigm for UAV safety, emphasizing the need for innovative approaches^[2]. Patel et al. (2017) delve into the design and analysis of UAV propellers, a critical aspect of safety^[26]. Murphy et al. (2008) investigate crew roles and operational protocols for micro-UAVs in urban environments, highlighting the importance of operational safety^[20].

Barrett et al. and Bergevin address UAV safety and risk assessment from different perspectives. Barrett et al. focus on enhancing drone safety and productivity through sensor technologies^[4], while Bergevin discusses the challenges of testing manned and unmanned aerial vehicles, emphasizing the importance of rigorous safety protocols during testing phases^[5].

Hirling^[14] and Namian et al.^[21] contribute to this theme by discussing operational safety considerations and safety risks associated with UAVs in construction, respectively. These studies highlight the need for comprehensive risk assessment frameworks and regulatory compliance to ensure safe UAV operations.

B. Propeller Design and Safety

Several studies focus on propeller design and its implications for safety. Nguyen et al. (2020) work towards the design of deformable propellers for drone safety, addressing the issue of propeller-related injuries^[23]. Balestrieri et al. (2021) provide an overview of sensors and measurements for UAV safety, considering the role of sensors in enhancing safety^[3]. Wang et

al. (2016) discuss risk considerations in the use of UAVs in the construction industry, shedding light on the practical safety challenges^[34].

Bonci et al. and Budiyono and Iwata explore propeller design solutions aimed at enhancing UAV safety. Bonci et al. propose the Double Propeller Ducted-Fan UAV for safe infrastructure inspection^[6], while Budiyono and Iwata delve into safety considerations in the design and testing of rotary-wing UAVs^[8].

Tracy focuses specifically on propeller design and analysis for small autonomous UAVs, emphasizing aerodynamic performance and structural integrity to ensure safe propeller operation^[32].

C. Experimental Evaluation of Safety Measures

A significant portion of the literature is dedicated to experimental evaluations and safety testing methodologies. Igarashi et al. (2021) evaluate the effectiveness of protective glasses for small UAV propellers through preliminary experiments^[15]. Denney and Pai (2016) explore safety considerations for UAS ground-based detect and avoid systems, emphasizing the need for real-world testing^[11].

Franke et al. and Raja et al. contribute to this theme by conducting experimental evaluations of safety measures related to UAVs. Franke et al. examine impact scenarios and consequences of collisions involving UAVs^[12], while Raja et al. investigate asymmetrical damage aspects of disc brakes in long-range UAVs^[28].

D. Modeling and Simulation for Safety Assessment

Some researchers have employed modeling and simulation techniques to assess safety aspects. Rattanagraikanakorn et al. (2020) use multibody system modeling to study UAV collisions with the human head, contributing to a deeper understanding of the consequences of such collisions^[29]. Schmidt and Pretschner (2022) present StellaUAV, a tool for testing the safe behavior of UAVs using scenario-based testing^[30].

Bui et al. and Guo et al. utilize modeling and simulation techniques to assess UAV safety. Bui et al. propose a bio-inspired deformable propeller structure^[9], while Guo et al. present safety control mechanisms for quadrotor UAVs, addressing ground effect and blade damage through simulation-based approaches^[13].

E. Safety Regulations and Standards

The literature also addresses the regulatory and standardization aspects of UAV safety. Lee et al. (2022) discuss safety and privacy regulations for UAVs, providing a comparative analysis of regulatory frameworks^[18]. Cawthorne and Devos (2020) introduce the concept of "capability caution" in UAV design, emphasizing the need for cautious design to enhance safety^[10].

Pető and Uhlig et al. focus on safety regulations and standards governing UAV operations. Pető provides a comprehensive review of safety and security issues of UAVs^[27], while Uhlig et al. discuss safety and reliability considerations within UAV construction, emphasizing the importance of adherence to regulatory frameworks and quality control measures^[33].

This comprehensive literature review provides valuable insights into the multifaceted landscape of UAV propeller safety, ranging from design considerations to experimental evaluations and regulatory frameworks. These diverse perspectives contribute to a deeper understanding of the challenges and opportunities in ensuring the safe operation of UAVs in various domains.

III. METHODOLOGY

A. Test Setup and Equipment

The crash test of a propeller of about 30 inches was conducted to evaluate the harm caused by the impact of the propeller to human.

A propeller mounted on a drone is considered to cause significant harm when it comes into direct contact with a person. It is necessary to develop a test method to evaluate the cut caused by the propeller when the drone collides with the head or other parts of the body. In this test, a test method will be developed to reproduce the case of a propeller impact of about 30 inches at a low speed of up to 20 km/h. A crash test was conducted and various data was obtained.

B. Experimental Design

Figure 1 presents a schematic diagram of the test apparatus, offering a visual representation of the setup used in our research to evaluate the harm caused by propeller impacts. This

apparatus is designed to replicate real-world scenarios where a UAV propeller collides with a human, such as the head or other body parts. The diagram illustrates the key components involved in the testing process, including the UAV propeller, the human surrogate target, and the controlled testing environment.

Figure 2 provides an overview of the overall test apparatus, offering a comprehensive visual representation of the experimental setup used in our research. This figure encapsulates the key components and their spatial relationships, providing context for the subsequent detailed illustrations.

Figure 3 delves into the bogie and traction device, presenting both front and side perspectives. These components play a crucial role in guiding and controlling the trajectory of the collision tests, ensuring precise and reproducible test conditions.

Figure 4 offers an up-close look at the stoppers and the dummy doll head, which serve as the primary collision targets. The stoppers' strategic placement restricts the surrogate's motion, mimicking real-world scenarios, while the dummy doll head represents the point of impact, facilitating the assessment of harm caused by propeller collisions.

Figure 5 captures a moment from a single propeller collision test, providing a dynamic view of the experiment in action. This illustration visually encapsulates the critical instance when the UAV propeller makes contact with the surrogate target, offering insight into the test's execution and the data collection process.

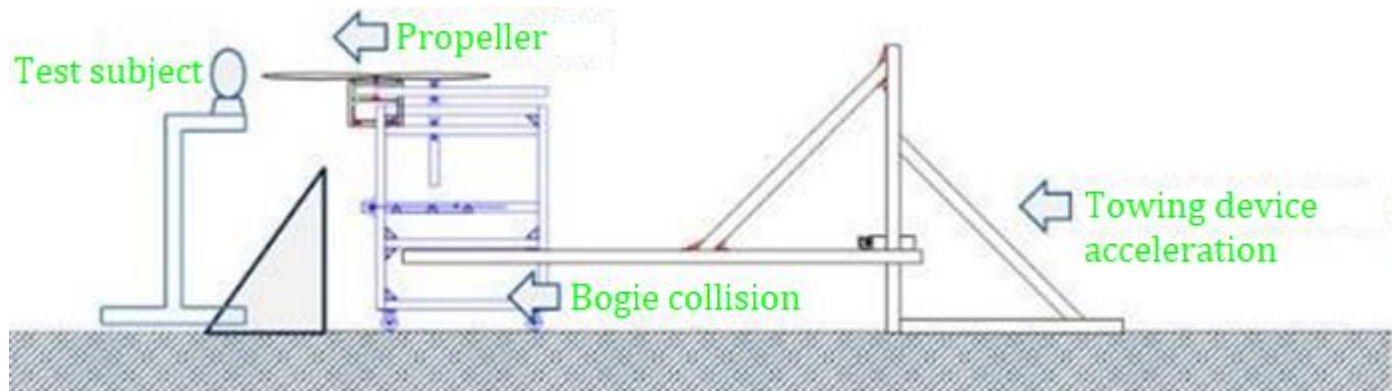


Figure 1 Schematic diagram of the test apparatus



Figure 2 Overall test apparatus



Figure 3 Bogie and traction device (front, side)

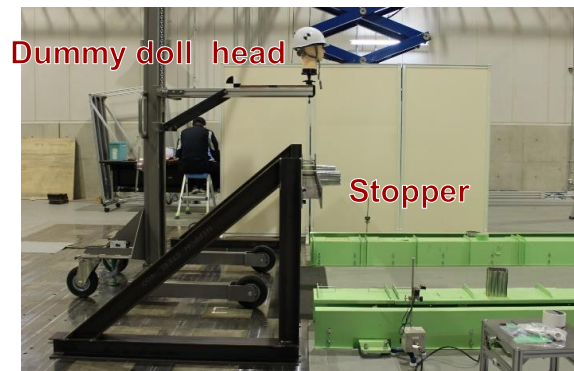


Figure 4 Stoppers and dummy doll head to be collided with



Figure 5 Single propeller collision test

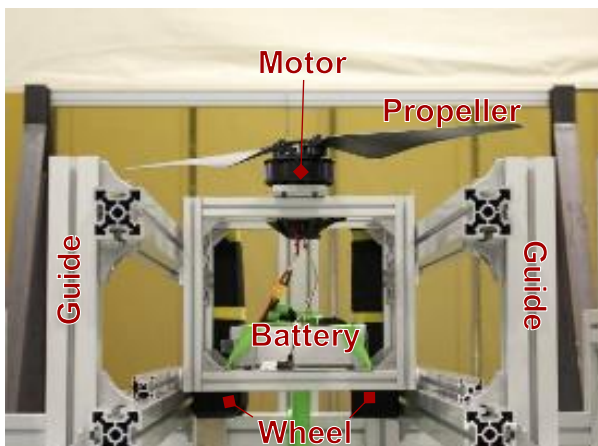


Figure 6 Configuration of a single propeller



Figure 7 Four types of propeller

Figure 6 details the configuration of a single propeller, offering a close-up view of this essential component of our research setup. This illustration provides a clear and comprehensive depiction of the propeller's design, structure, and orientation within the experimental apparatus. Understanding the precise configuration of the propeller is pivotal in comprehending how it interacts with the surrogate target during collision tests, thus contributing to a deeper understanding of the research methodology employed in our study.

C. Propellers and protective equipment

There are four types of propellers used in this test. The propeller specifications are shown in Table 1 and each propeller is shown in Figure 7. Each propeller is different in size and manufacturer. The motors and batteries used to rotate each propeller are different, so the overall weight of the propeller as a single unit is also different. Since each propeller has two different directions of rotation (CW and CCW), the direction of rotation of the motor was also set according to the propeller.

The subjects were two types of head mannequins (TYPE A and B) shown in Figure 8. The head mannequin consists of a rubber surface and a sponge interior. The forearm dummies consist of a simulated human bone and arm-hard silicone, and TYPE C also has a simulated blood vessel with red ink. The head mannequin and forearm dummies were used as the base for evaluation of cut wounds with several protective devices. Table 2 lists the protective equipment used, and Figure 9 shows each protective equipment worn on the base head mannequin and forearm dummy. Note that glasses 1 are secured with masking tape to prevent them from falling off. Since protective pants cannot be attached to the head mannequin and forearm dummy, they were attached to the human body dummy AM50 for automobile collision (hereinafter referred to as "dummy").

IV. PROPELLER TESTING AND DATA COLLECTION

A. Test Facilities and Equipment

Figure 10 shows the angle of view of the high-speed video cameras. Two high-speed video cameras were used to capture images in order to understand the behavior of a single propeller during a collision. The filming speeds were all set at 10,000 frames per second. As shown in Figure 11, a reflective marker was attached to the propeller to measure the rotation speed of the propeller unit, and a non-contact digital tachometer was used to measure the rotation speed. The rotation speed was measured before the start of each test.

B. Results of Propeller Impact Tests

Table 3 shows the list of the test results. The collision velocity is the value obtained by image analysis of the high-

speed video. The details of the cut conditions are described in Appendix.

In the condition with protective equipment, all the protective equipment, except for the helmet of test number 1, was destroyed or dislodged due to contact with the propeller, resulting in wounds on the base subject. Since the propeller made contact multiple times while rotating, even if the Protective equipment could withstand the first contact, it was destroyed by subsequent contacts.

No differences were observed in the cut conditions at different impact speeds when comparing test numbers 12 and 13. It is considered that the effect of impact speed was small because the rotational speed of the propeller was sufficiently fast.

C. Interpretation of Findings

The finding obtained from this test is that the larger the diameter of the propeller, the deeper the depth of the cut. In the future, we will measure the depth of the cut wound to clarify the correlation with the size of the propeller diameter, thereby accumulating data useful for risk assessment.

Figure 12 and Figure 13 show the behavior of test numbers 7 and 9 during a crash as representative of the two conditions without and with Protective equipment. Others are shown in the Appendix. Note that test number 25, in which the high-speed video camera was unable to record images due to equipment trouble, is excluded.

Figure 14 to Figure 38 show the cut conditions of the subject, propeller, and protective equipment.

V. DISCUSSION

The cut wounds caused when the subject and the propeller made contact were identified as deep wounds, as in Test No. 8, and contact marks that stroked the table surface, as in Test No. 20. In the former case, the propeller was hard, and even though the tip was damaged at the first contact, the remaining propeller caused multiple deep wounds. In the latter, after the tip was damaged by the first contact, the propeller deflected, and although there were no deep scratches, there were numerous contact marks.

As shown in Figure 39, the fixture for the head mannequin was made of plastic, and the propeller blows sometimes moved the position of the head mannequin or dislodged it from the fixture. Therefore, a more rigid fixation method could make the cut conditions more severe. As shown in Figure 39, the forearm dummy was sandwiched between metal plates and fixed to a wooden stand with screws, so the forearm dummy did not move even after multiple blows from the propeller.

Table 1 Propeller specifications

Name	T27	H30	T32	D33
Propeller size	27inch	30inch	32.2inch	33inch
Propeller maker	T-Motor	HOBBYWING	T-Moter	DJI
Weight [kg]	10.2	11.8	13.8	10.8

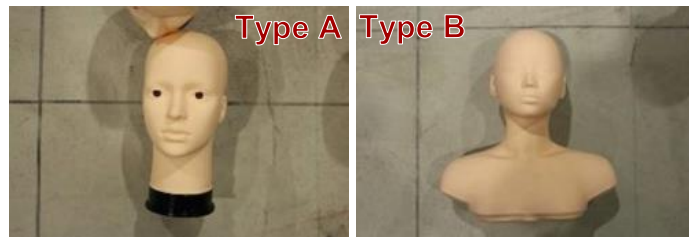


Figure 8 Two types of dummy doll heads

Table 2 List of protective equipment

Abbreviation	Kind	Product name
Helmet	Helmet	Monotaro Helmet MPtype SS-101
Glasses1	Protective glasses	Daiso Safety glasses K290
Glasses2	Protective glasses	Monotaro Safety goggles Antifog type
Glasses3	Protective glasses	TRUSCO Safety goggles TSG-501T
FG1	Face guard	TRUSCO BM-HFC (ANSI Z87.1)
FG2	Face guard	Monotaro face guard
FG3	Face guard	TRUSCO face guard BM-PC

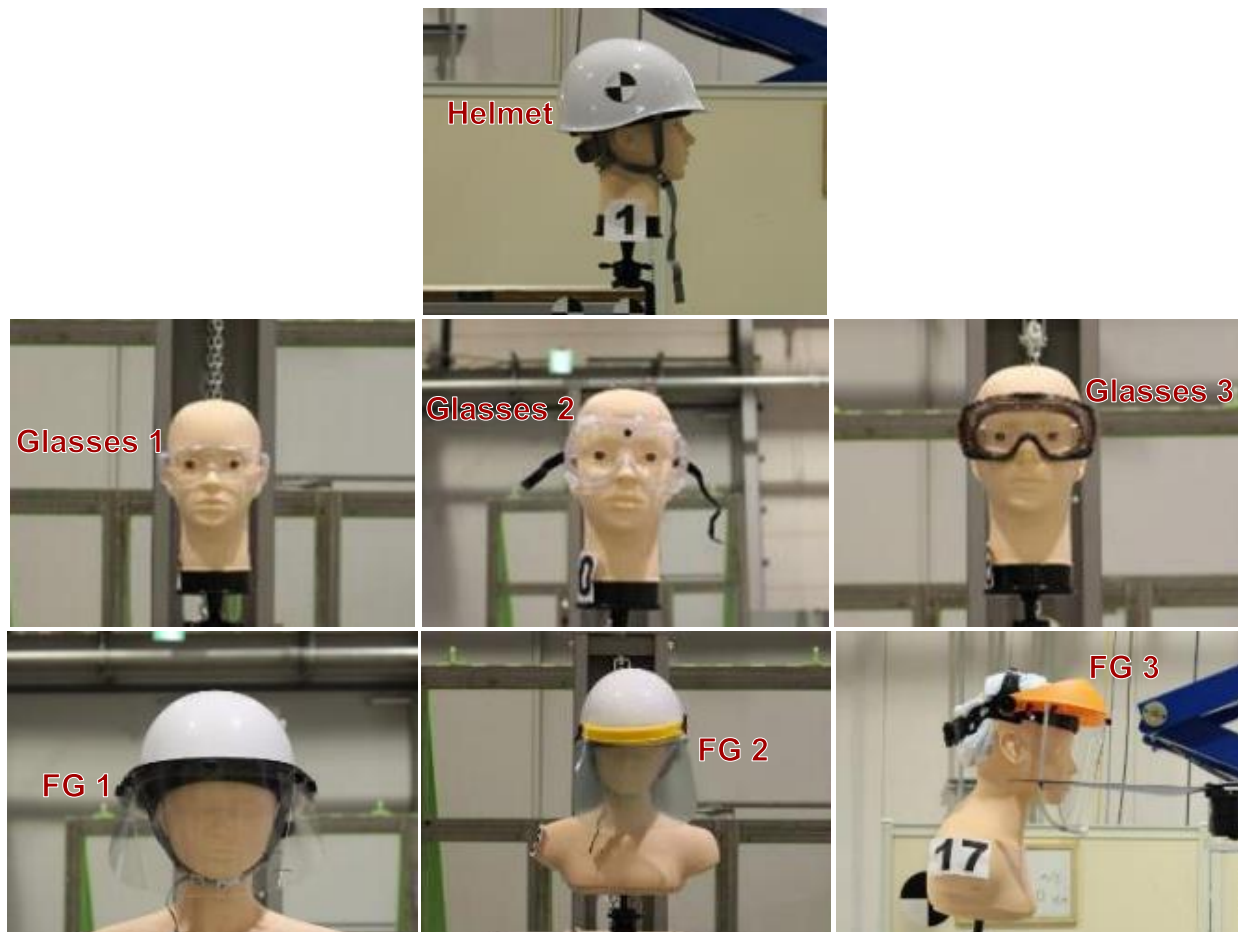


Figure 9 Protective equipment

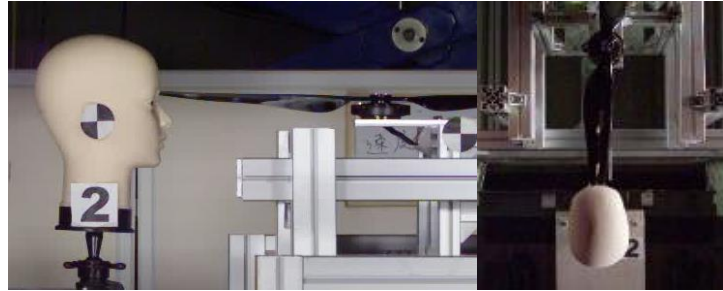


Figure 10 Angle of view of high-speed video

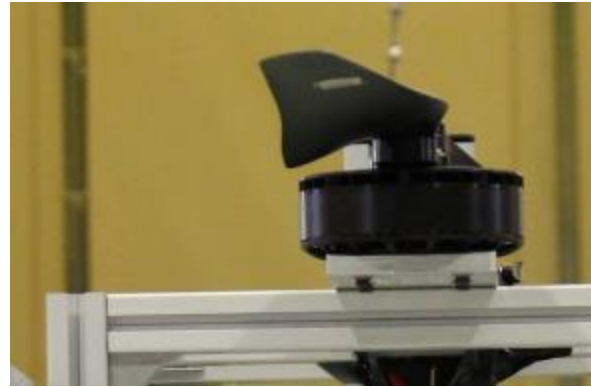


Figure 11 Reflective marker attached to propeller

Table 3 List of test results

Test No.	target	dummy	Protect	Collision speed (m/s)	Propeller			Cut Wound	
					Type	RPM	CW/CCW	Protect	dummy
1	Buccal	TYPE A	Helmet	2.0	T27	2880	CCW	Yes	No
2	Buccal	TYPE A	None	2.0	T27	2902	CCW	Yes	Yes
3	Buccal	TYPE A	None	2.0	T27	2904	CCW	Yes	Yes
4	Buccal	TYPE A	Glasses1	1.9	T27	2892	CCW	Yes	Yes
5	Buccal	TYPE A	Glasses1	1.9	T27	3850	CCW	Yes	Yes
6	Buccal	TYPE A	None	2.0	H30	3960	CW	Yes	Deep
7	Buccal	TYPE A	None	1.8	H30	4060	CW	Yes	Deep
8	Buccal	TYPE A	None	1.9	H30	4047	CW	Yes	Deep
9	Buccal	TYPE A	Glasses1	1.9	H30	4001	CW	Yes	Deep
10	Buccal	TYPE A	Glasses2	1.9	H30	4024	CW	Yes	Deep
11	Buccal	TYPE A	Glasses2	5.1	H30	3935	CW	Yes	Deep
12	Buccal	TYPE A	None	5.0	H30	4092	CW	Yes	Deep
13	Buccal	TYPE A	None	1.7	H30	4076	CW	Yes	Deep
14	Buccal	TYPE A	None	5.0	H30	4042	CW	Yes	Deep
15	Buccal	TYPE B	FG1	1.9	H30	4006	CW	Yes	Yes
16	Buccal	TYPE B	FG2	1.8	H30	3992	CW	Yes	Deep
17	Buccal	TYPE B	FG3	1.9	H30	3945	CW	Yes	Deep
18	Buccal	TYPE A	None	1.8	T32	4601	CCW	Yes	Deep
19	Buccal	TYPE A	Glasses2	1.9	T32	4567	CW	Yes	Yes
20	Buccal	TYPE A	Glasses3	1.9	T32	4525	CW	Yes	Yes
21	Buccal	TYPE B	FG1	1.9	T32	4503	CW	Yes	Yes
-	-	-	-	-	-	-	-	-	-
23	Buccal	TYPE B	FG1	1.8	D33	3229	CCW	Yes	Deep
24	Buccal	TYPE A	None	1.4	D33	2942	CCW	Yes	Deep
25	Buccal	TYPE A	Glasses2	1.4	D33	2884	CCW	Yes	Deep
26	Buccal	TYPE A	Glasses3	1.5	D33	3027	CCW	Yes	Deep

A. Analysis of Test Results

The results of our propeller collision tests, as illustrated in Figure 1, provide critical insights into the potential harm caused by UAV propeller impacts on human surrogates. These tests were conducted under controlled conditions to replicate real-world scenarios, both with and without protective equipment. Notably, Test No. 25 was excluded due to equipment issues. Our observations revealed two distinct categories of injuries resulting from these collisions: deep wounds, as observed in Test No. 8, and contact marks, as seen in Test No. 20. In cases of deep wounds, the propeller's hardness led to multiple significant lacerations even after initial damage to the propeller tip. Conversely, contact marks were characterized by deflection upon initial contact, resulting in numerous surface-level marks.

B. Implications of Propeller Variability

Figure 39 highlights the influence of propeller variability on test outcomes. The fixture for the head mannequin, constructed from plastic, occasionally shifted or dislodged due to the force of propeller blows. This variability in fixture rigidity potentially influenced the severity of cut wounds. To further refine the testing methodology and ensure consistent results, future research should explore more robust fixation methods to stabilize the head mannequin during propeller impacts.

C. Protective Gear Effectiveness

The efficacy of protective gear, as depicted in Figure 12 and Figure 13, emerges as a crucial consideration. The presence of protective equipment significantly altered the injury outcomes during propeller collisions. These findings underscore the importance of protective gear in mitigating harm caused by propeller impacts. Further research and development in this area hold promise for enhancing the safety of UAV operators and bystanders.

D. Safety Considerations for UAV Operators

Our study's findings carry direct implications for UAV operators, emphasizing the paramount importance of safety measures during drone operations. It is imperative for operators to be aware of the potential harm that propellers can inflict and to prioritize safety protocols, including the use of protective gear and adherence to operational guidelines.

E. Comparison with Existing Research

In comparing our research with existing studies, our work contributes valuable data on the consequences of propeller-human interactions. While prior research has addressed various aspects of UAV safety, our focus on injury assessment during propeller impacts complements the broader safety landscape.

F. Limitations of the Study

It is essential to acknowledge the limitations of this study. Test No. 25's exclusion due to equipment issues highlights the need for rigorous equipment maintenance and robust data collection procedures. Additionally, the observed variability in fixture rigidity calls for further investigation into fixture design improvements.

In conclusion, our research offers crucial insights into the safety considerations surrounding rotary-wing UAV propeller

testing. The analysis of test results, implications of propeller variability, and the effectiveness of protective gear underscore the importance of continued research in this field. These findings contribute to safer UAV operations and serve as a foundation for future advancements in UAV safety standards and protective measures.

VI. CONCLUSION

In summary, this study has provided valuable contributions to the realm of UAV safety, specifically focusing on rotary-wing UAV propeller testing and injury assessment. Our research has unveiled critical insights that have broad implications for the safety of UAV operators and bystanders in various operational scenarios.

A. Summary of Key Findings

The analysis of propeller collision tests, conducted both with and without protective equipment, has yielded significant findings. We observed two distinct types of injuries resulting from these collisions: deep wounds, characterized by multiple lacerations caused by rigid propellers, and contact marks, resulting from propeller deflection upon initial contact. These observations shed light on the varying injury profiles associated with propeller impacts, highlighting the potential risks that must be addressed to ensure safety.

B. Contributions to UAV Safety

Our study makes noteworthy contributions to the field of UAV safety. First and foremost, it emphasizes the pivotal role of protective gear in mitigating the harm caused by propeller impacts. The presence of protective equipment significantly altered injury outcomes, underlining the importance of its utilization in UAV operations. Furthermore, our research highlights the need for robust design standards and testing protocols to enhance propeller safety. The findings also underscore the significance of fixture rigidity in testing procedures, suggesting that future research should explore improved fixation methods for consistent results.

C. Future Research Directions

As we move forward, there are several promising avenues for future research in the domain of UAV safety. Investigating advanced protective gear designs and materials could lead to more effective injury prevention measures. Additionally, refining fixture designs and testing methodologies to ensure stability during propeller collisions remains an important research direction. Furthermore, studies assessing the impact of propeller variability on injury outcomes and developing standardized safety guidelines for UAV operators are crucial for advancing the field.

In conclusion, this study represents a significant step towards enhancing the safety of rotary-wing UAV operations. The knowledge gained from our research can inform the development of improved safety protocols and protective measures, ultimately contributing to the responsible integration of UAVs into a wide array of applications and environments. As we continue to explore these research directions, we aim to foster a safer future for UAV operators, bystanders, and the broader community.

VII. ACKNOWLEDGMENTS

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Figure 12 Behavior of test number 7 during collision
(without protective equipment, 1.9 m/s, H30)

Figure 13 Behavior of Test No. 10 during collision
(Glasses2, 1.9 m/s, H30)



Figure 14 Damage to test number 1 (helmet, 2.0 m/s, T27)

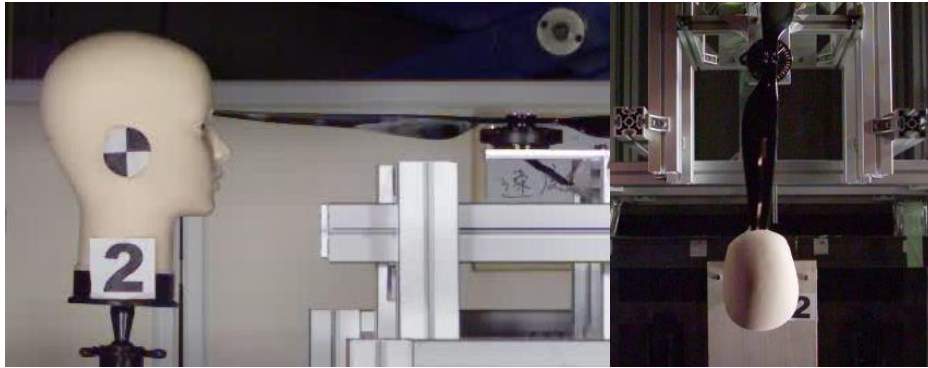


Figure 15 Damage in test number 2 (without protective equipment, 2.0 m/s, T27)



Figure 16 Damage in test no. 3 (without protective equipment, 2.0 m/s, T27)

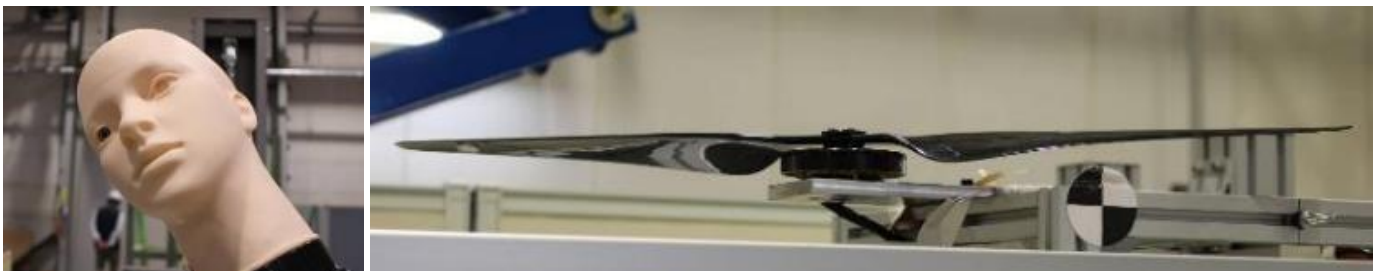


Figure 17 Damage of test number 4 (Glasses1, 1.9 m/s, T27)



Figure 18 Damage in test number 5 (Glasses1, 1.9 m/s, T27)



Figure 19 Damage in Test No. 6 (without protective equipment, 2.0 m/s, H30)

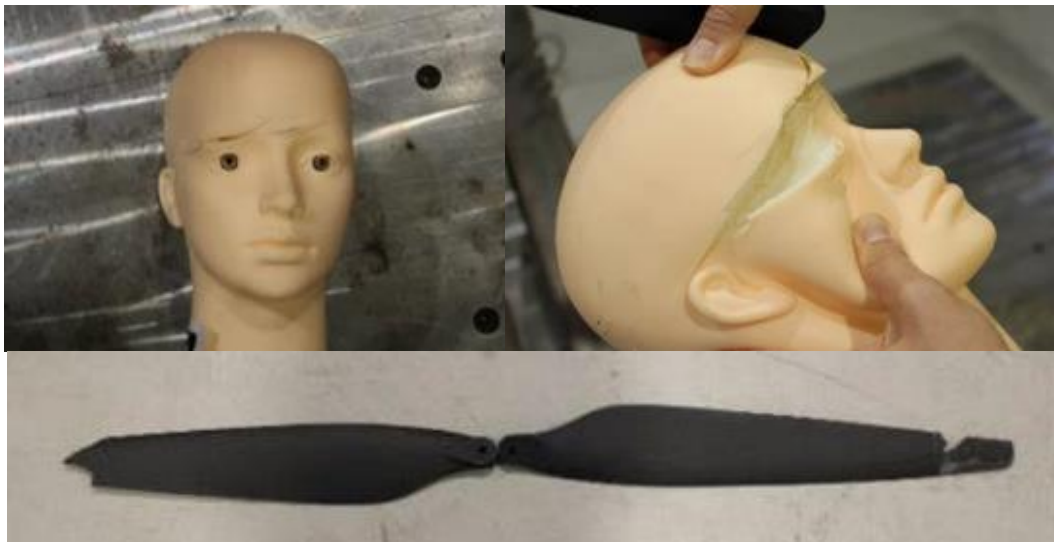


Figure 20 Damage of test number 7 (without protective equipment, 1.8 m/s, H30)

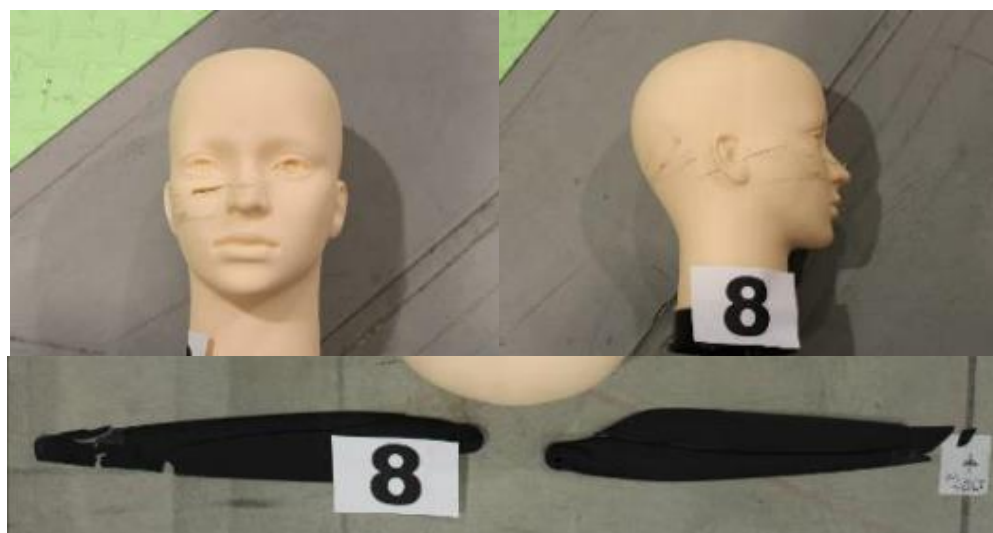


Figure 21 Damage in Test No. 8 (without protective equipment, 1.9 m/s, H30)



Figure 22 Damage to test number 9 (Glasses1, 1.9 m/s, H30)



Figure 23 Damage in Test No. 10 (Glasses2, 1.9 m/s, H30)

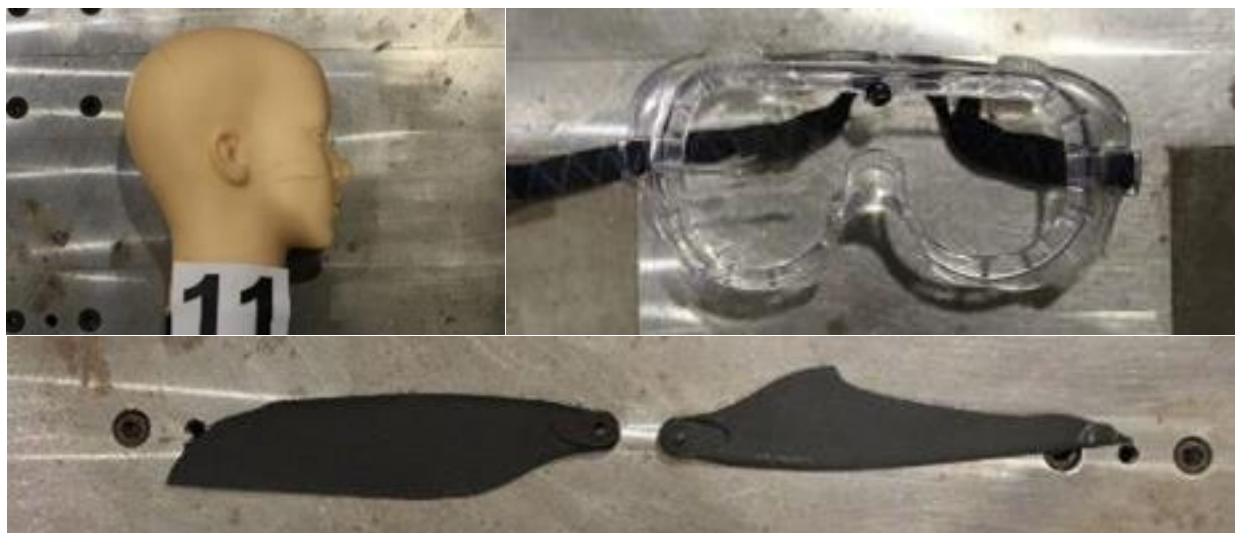


Figure 24 Damage to test number 11 (Glasses2, 5.1 m/s, H30)



Figure 25 Damage of Test No. 12 (without protective equipment, 5.0 m/s, H30)

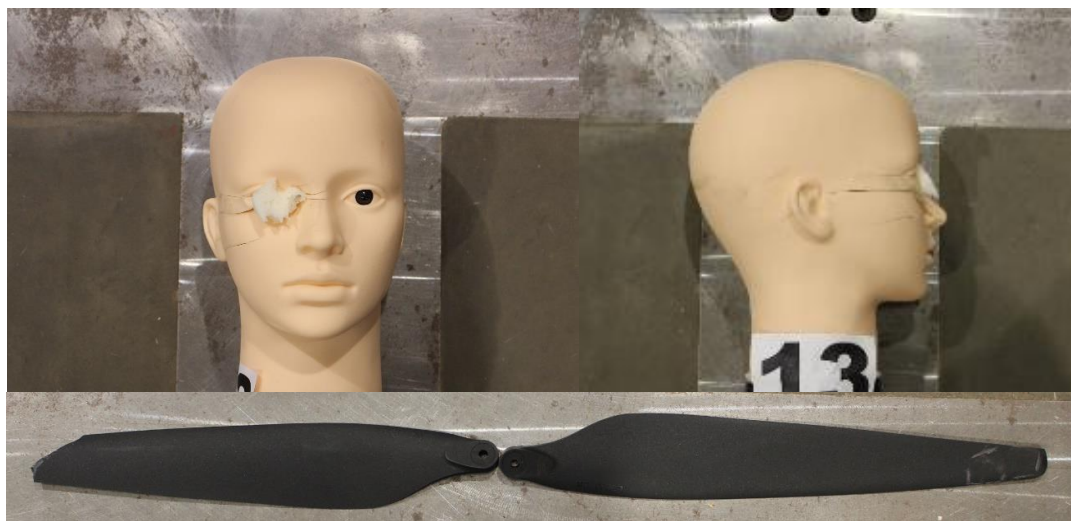


Figure 26 Damage of Test No. 13 (without Protective equipment, 1.7 m/s, H30)



Figure 27 Damage to Test No. 14 (without protective equipment, 5.0 m/s, H30)



Figure 28 Damage of Test No. 15 (FG1, 1.9 m/s, H30)



Figure 29 Damage to test number 16 (FG2, 1.8 m/s, H30)



Figure 30 Damage to Test No. 17 (FG3, 1.9 m/s, H30)



Figure 31 Damage in test no. 18 (without protective equipment, 1.8 m/s, T32)



Figure 32 Damage in Test No. 19 (Glasses2, 1.9 m/s, T32)

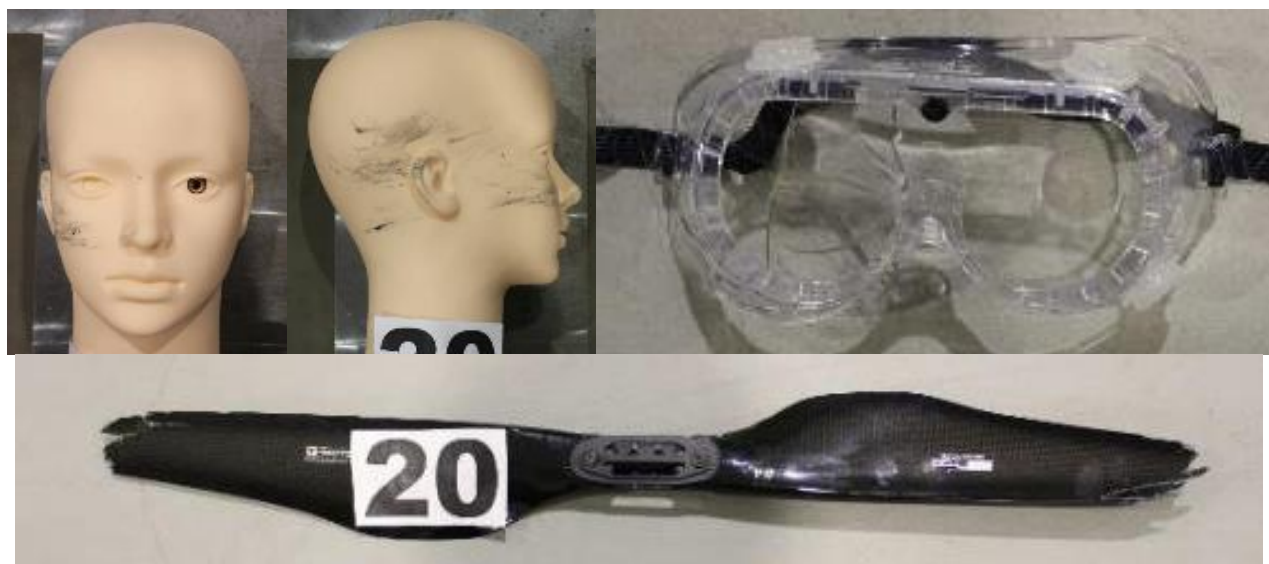


Figure 33 Damage in Test No. 20 (Glasses3, 1.9 m/s, T32)



Figure 34 Damage in test no. 21 (FG1, 1.9 m/s, T32)



Figure 35 Damage to test number 23 (FG1, 1.8 m/s, D33)



Figure 36 Damage in test no. 24 (without protective equipment, 1.4 m/s, D33)



Figure 37 Damage to test number 25 (Glasses2, speed measurement NG, D33)



Figure 38 Damage in Test No. 26 (Glasses3, 1.5 m/s, D33)



Figure 39 Fixing the head mannequin

APPENDIX

BEHAVIOR DURING COLLISION IN PROPELLER COLLISION EXPERIMENTS

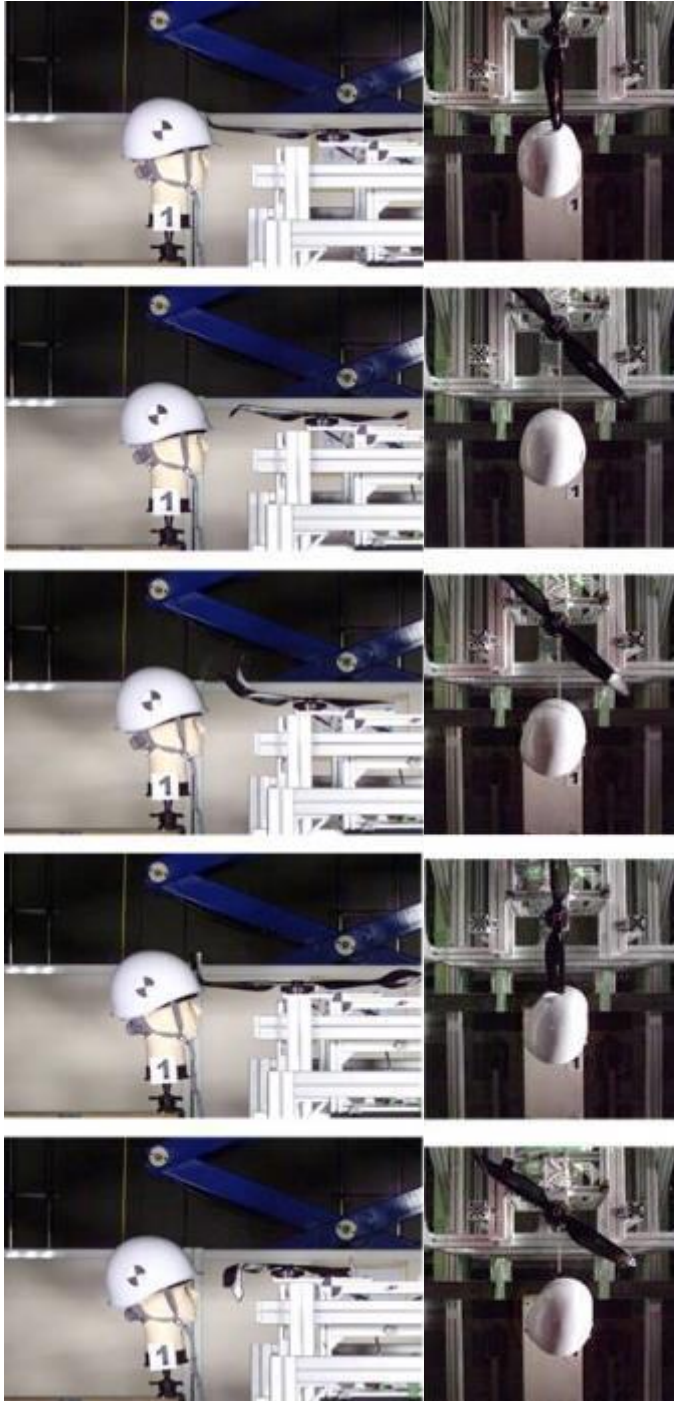


Figure A.1 Behavior of test number 1 during a crash
(helmet, 2.0 m/s, T27)

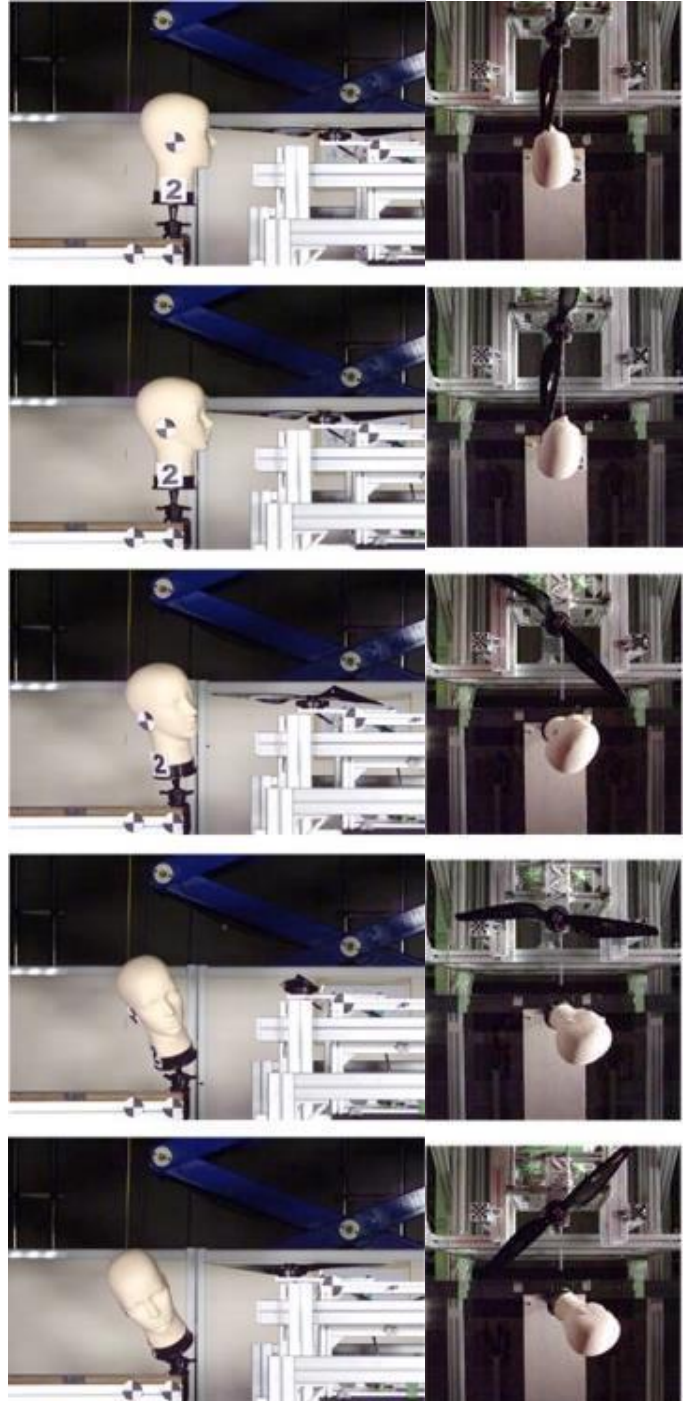


Figure A.2 Behavior of test number 2 during collision
(without protective equipment, 2.0 m/s, T27)

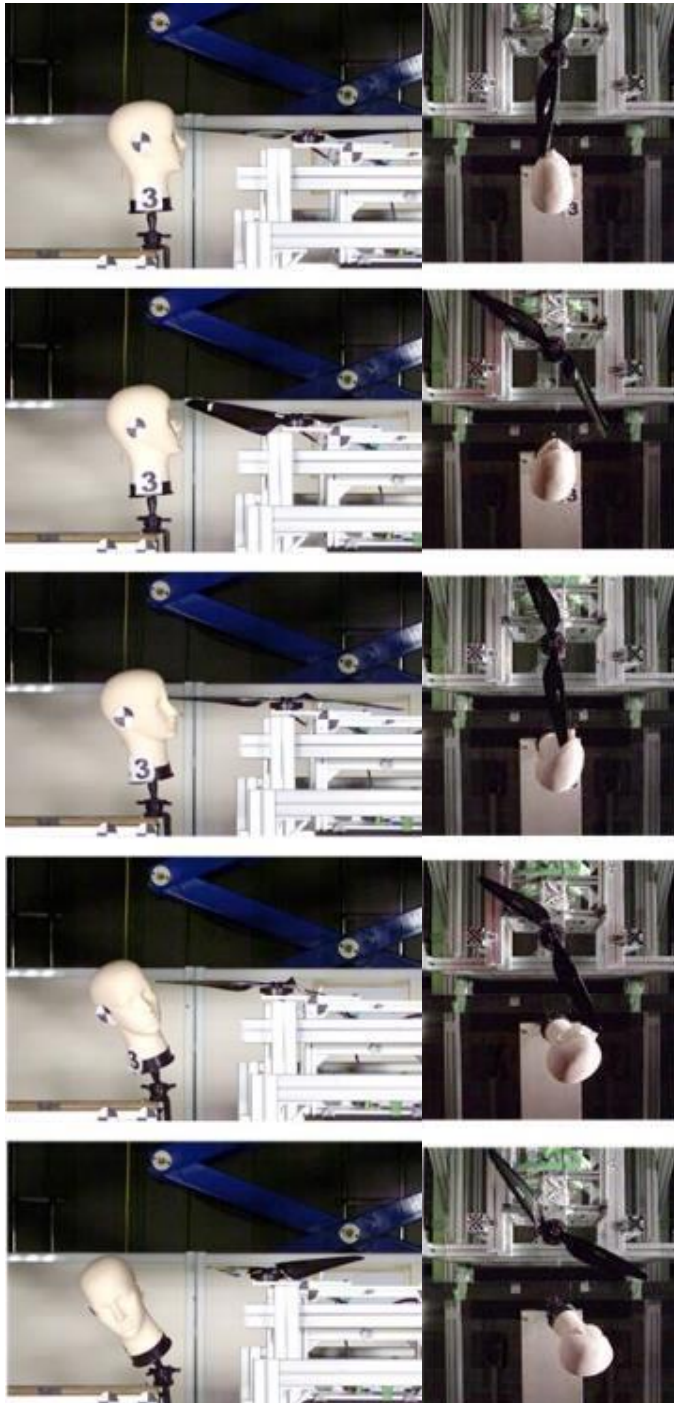


Figure A.3 Behavior of test number 3 during collision
(without protective equipment, 2.0 m/s, T27)

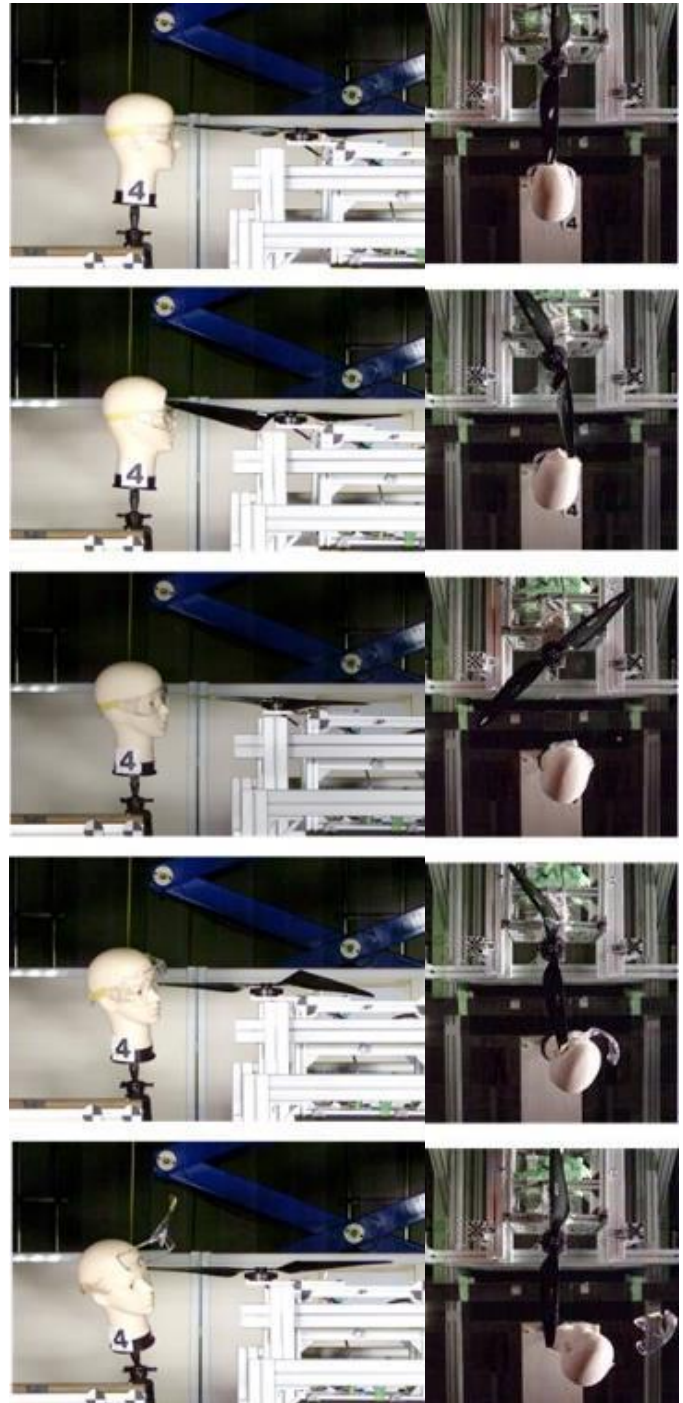


Figure A.4 Behavior of test number 4 during collision
(Glasses1, 1.9 m/s, T27)

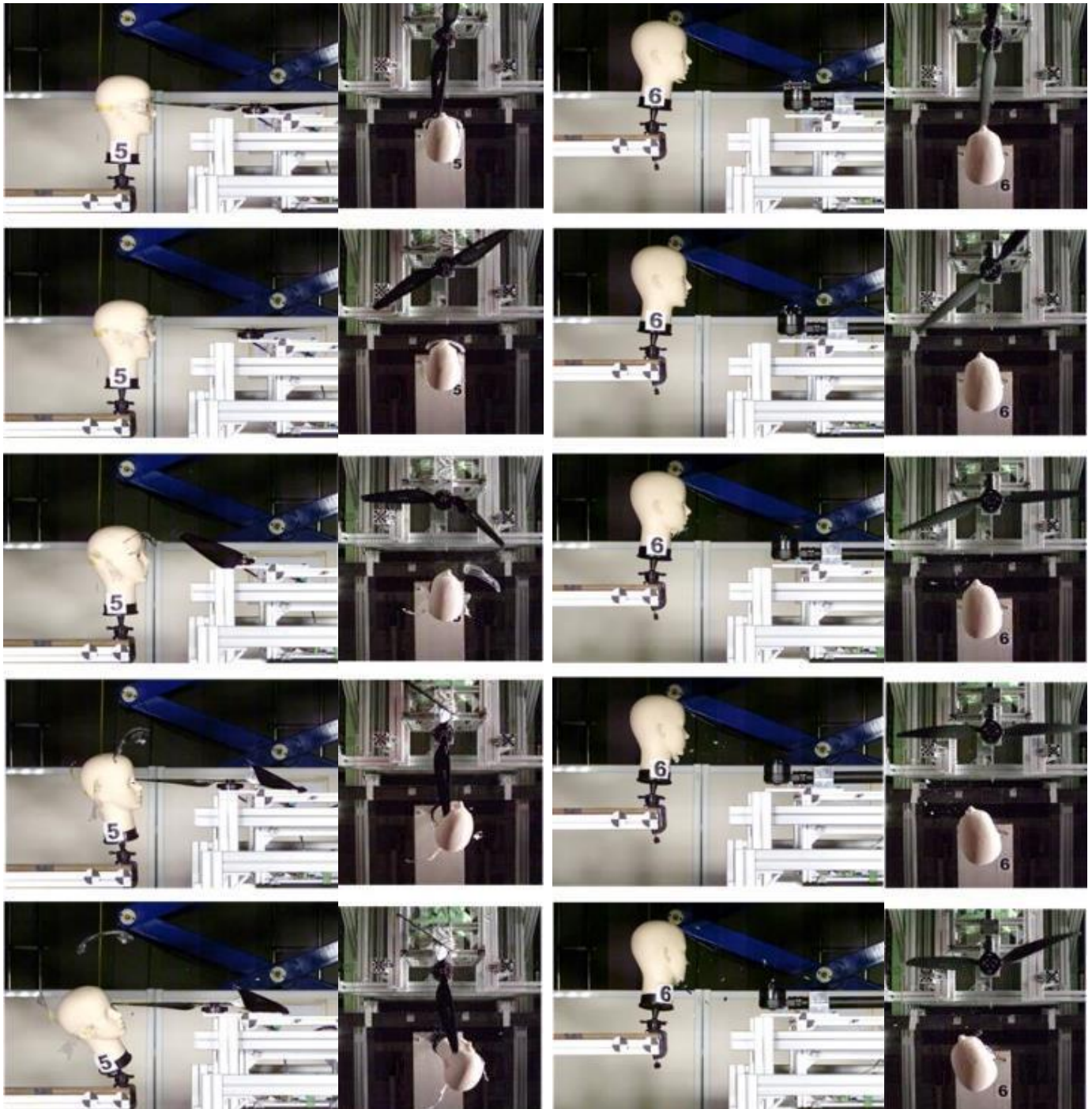


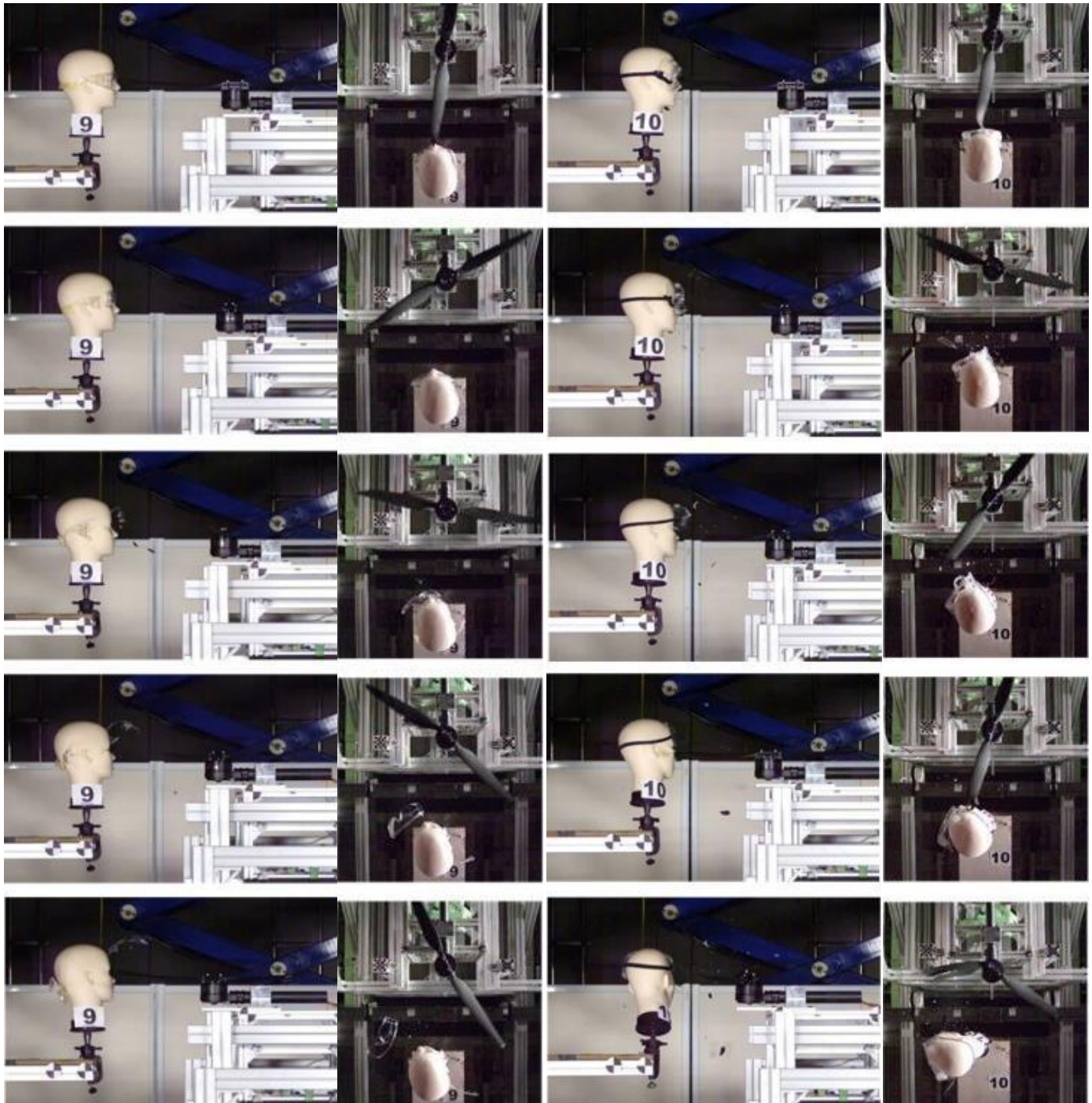
Figure A.5 Behavior of Test No. 5 during collision
(Glasses1, 1.9 m/s, T27)

Figure A.6 Behavior of Test No. 6 during collision
(without protective equipment, 2.0 m/s, H30)



Figure A.7 Behavior of test number 7 during collision
(without protective equipment, 1.8 m/s, H30)

Figure A.8 Behavior of test No. 8 during collision
(without protective equipment, 1.9 m/s, H30)



**Figure A.9 Behavior of Test No. 9 during collision
(Glasses1, 1.9 m/s, H30)**

**Figure A.10 Behavior of test No. 10 during collision
(Glasses2, 1.9 m/s, H30)**

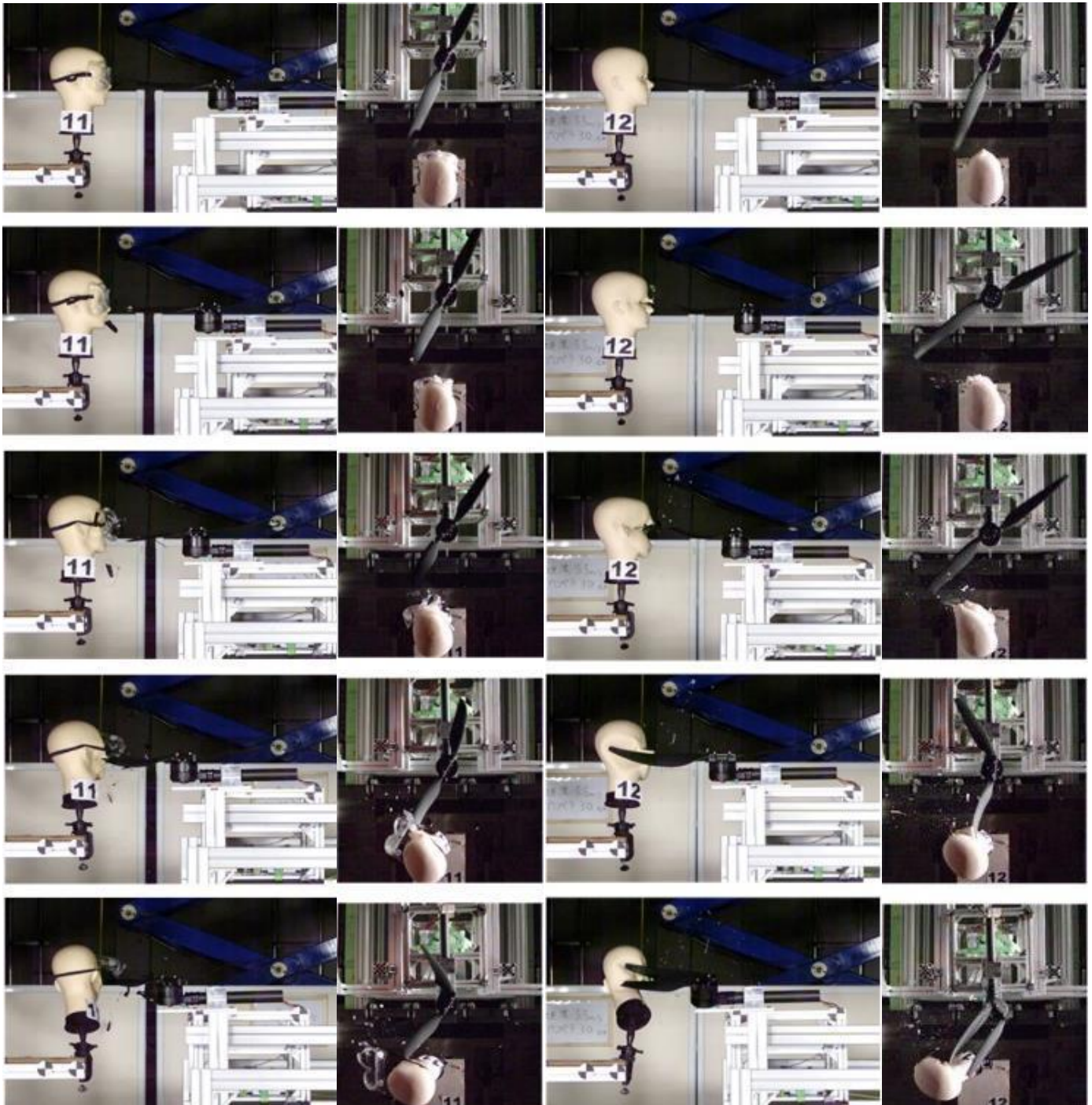


Figure A.11 Behavior of test number 11 during collision
(Glasses2, 5.1 m/s, H30)

Figure A.12 Behavior of Test No. 12 during collision
(without protective equipment, 5.0 m/s, H30)

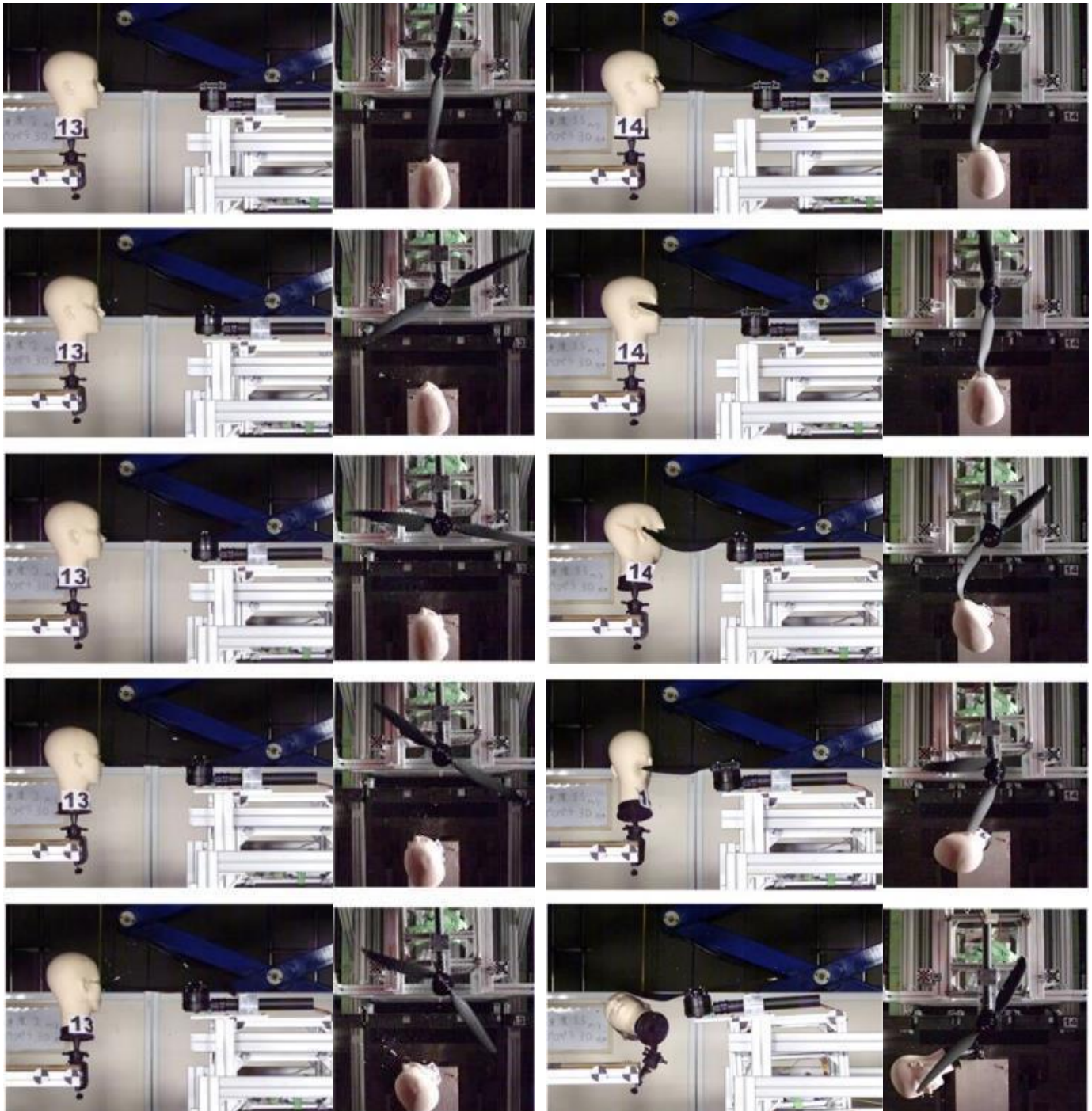


Figure A.13 Behavior of test No. 13 during collision
(without protective equipment, 1.7 m/s, H30)

Figure A.14 Behavior of test No. 14 during collision
(without protective equipment, 5.0 m/s, H30)



Figure A.15 Behavior of test No. 15 during impact
(FG1, 1.9 m/s, H30)

Figure A.16 Behavior of test number 16 during impact
(FG2, 1.8 m/s, H30)



Figure A.17 Behavior of test No. 17 during impact
(FG3, 1.9 m/s, H30)

Figure A.18 Behavior of test No. 18 during collision
(without protective equipment, 1.8 m/s, T32)



Figure A.19 Behavior of test number 19 during collision
(Glasses2, 1.9 m/s, T32)

Figure A.20 Behavior of Test No. 20 during collision
(Glasses3, 1.9 m/s, T32)

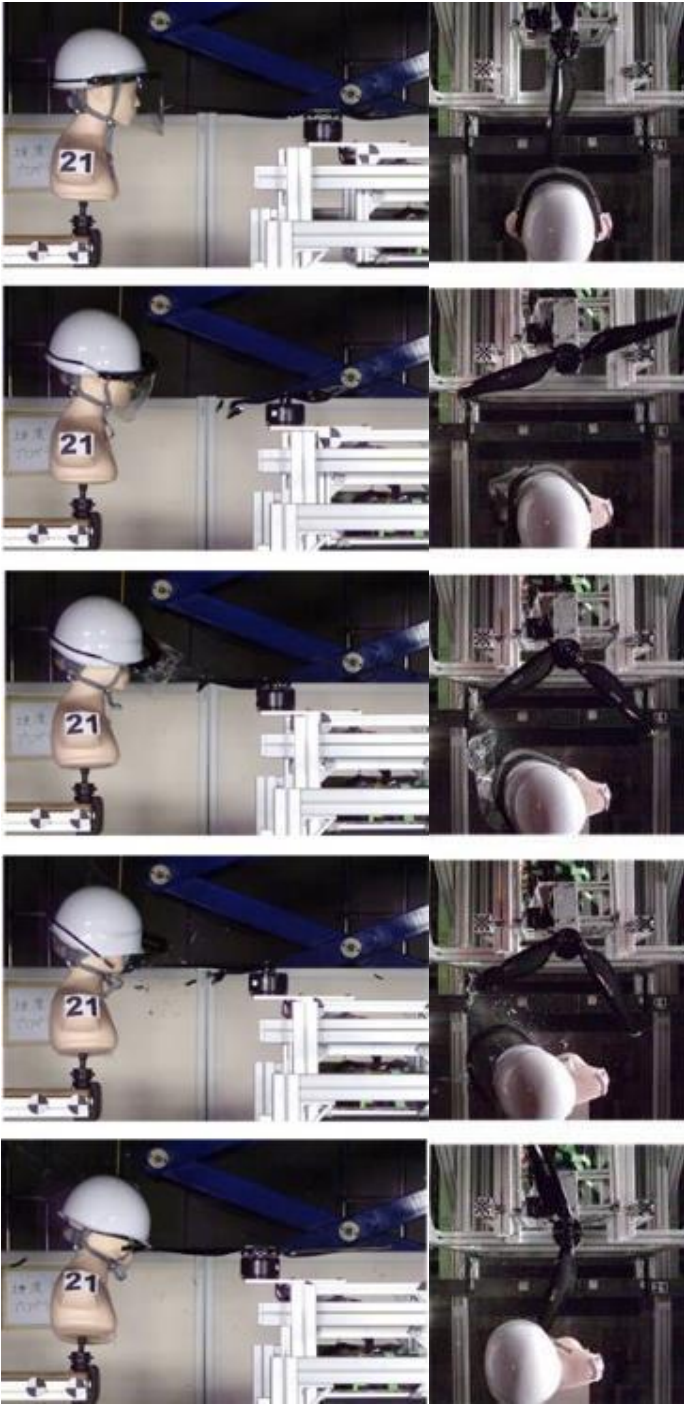


Figure A.21 Behavior of test No. 21 during impact
(FG1, 1.9 m/s, T32)

Figure A.22

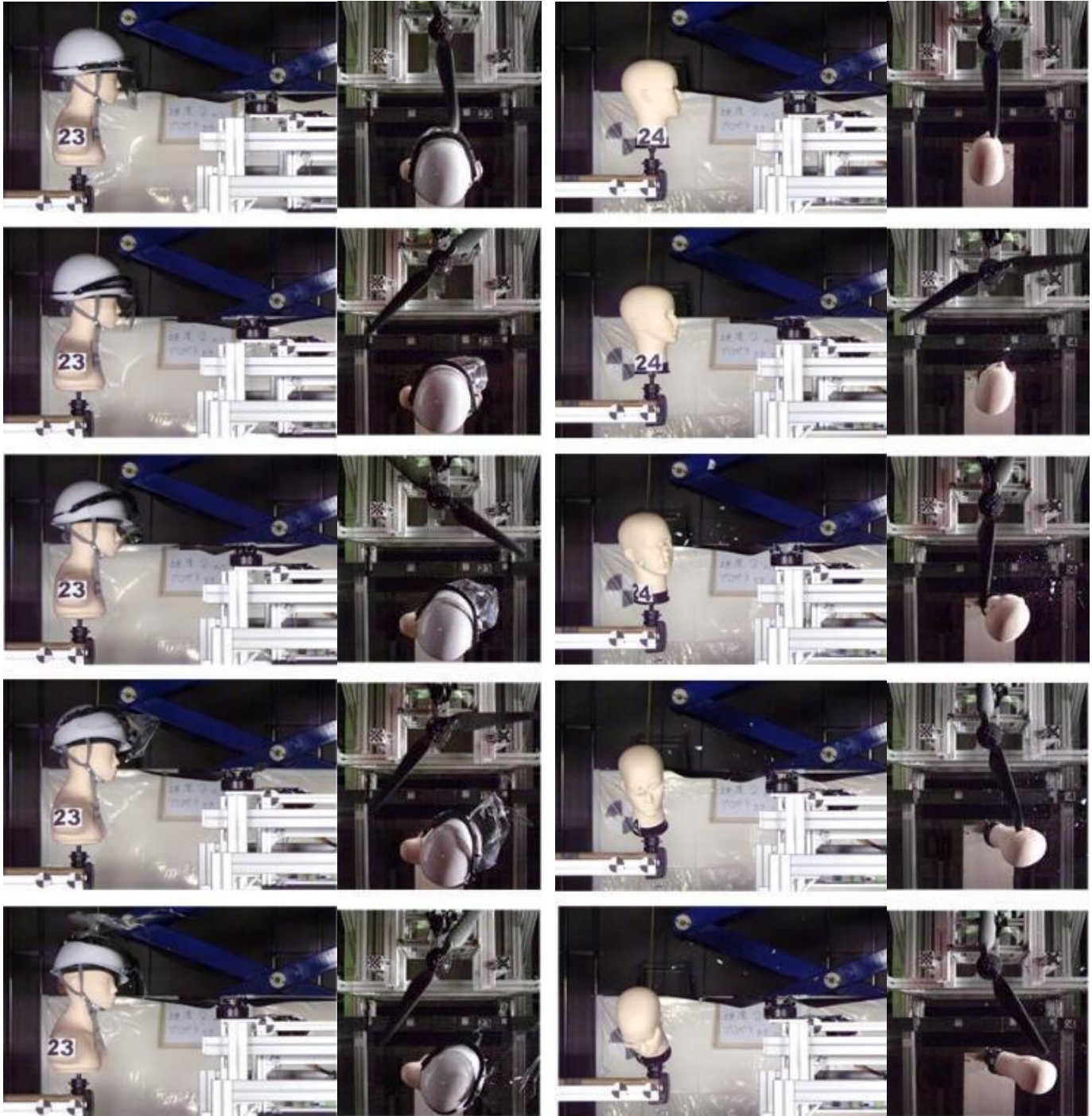


Figure A.23 Behavior of test number 23 during impact (FG1, 1.8 m/s, D33)

Figure A.24 Behavior of Test No. 24 during collision (without protective equipment, 1.4 m/s, D33)

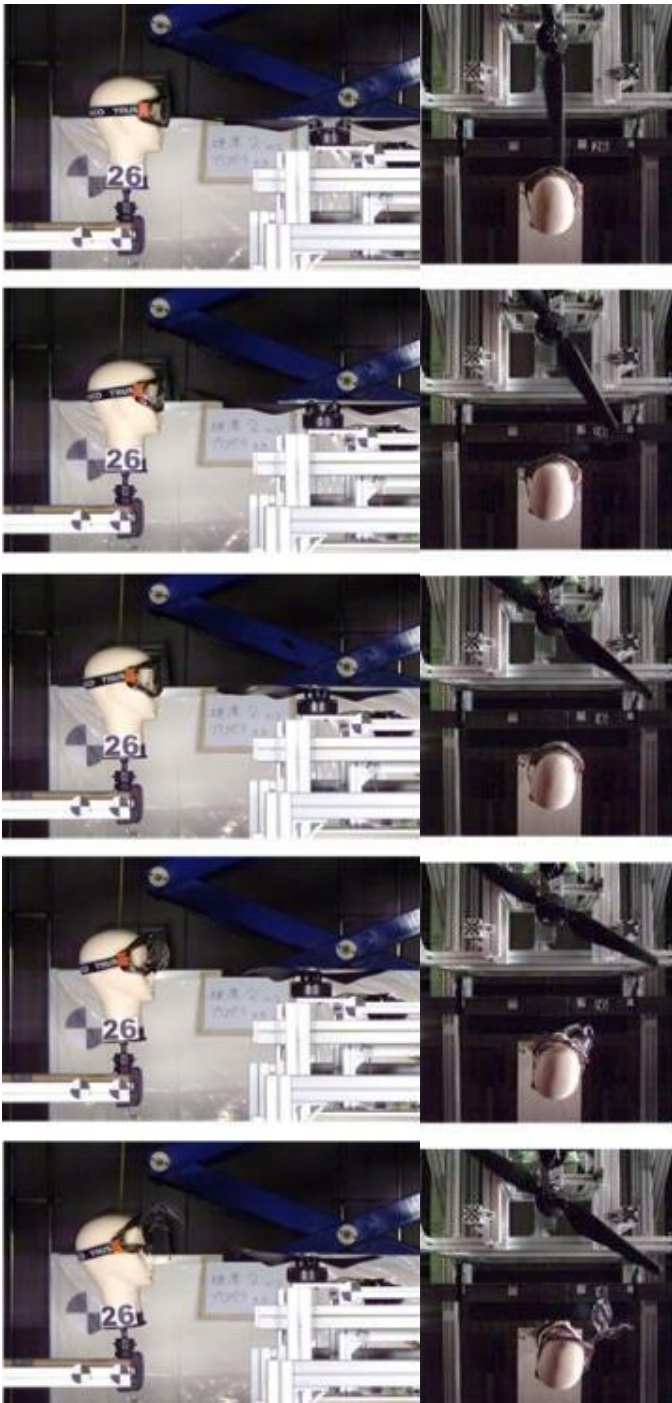


Figure A.25 Behavior of test number 26 during impact
(Glasses3, 1.5 m/s, D33)